

UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS

GCE Advanced Subsidiary Level and GCE Advanced Level

MARK SCHEME for the May/June 2007 question paper

9702 PHYSICS

9702/02

Paper 2 (AS Structured Questions), maximum raw mark 60

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began.

All Examiners are instructed that alternative correct answers and unexpected approaches in candidates' scripts must be given marks that fairly reflect the relevant knowledge and skills demonstrated.

Mark schemes must be read in conjunction with the question papers and the report on the examination.

- CIE will not enter into discussions or correspondence in connection with these mark schemes.

CIE is publishing the mark schemes for the May/June 2007 question papers for most IGCSE, GCE Advanced Level and Advanced Subsidiary Level syllabuses and some Ordinary Level syllabuses.



Page 2	Mark Scheme	Syllabus	Paper
	GCE A/AS LEVEL – May/June 2007	9702	2

- 1 (a) (i) all positions (accept 20, 40, 60, 80) marked to within $\pm 5^\circ$
positions are 40° , 70° , 90° and 102°
(-1 for each error or omission) B2
- (ii) allow $107^\circ \rightarrow 113^\circ$ B1 [3]
- (b) e.g. more sensitive at low volumes B1 [1]
(do not allow reference to 'accuracy')
- 2 (a) force per unit positive charge (on a small test charge) B1 [1]
- (b) field strength = $(210 / \{1.5 \times 10^{-2}\}) = 1.4 \times 10^4 \text{ N C}^{-1}$ A1 [1]
- (c) (i) acceleration = Eq / m C1
 $= (1.4 \times 10^4 \times 1.6 \times 10^{-19}) / (9.1 \times 10^{-31})$ C1
 $= 2.5 \times 10^{15} \text{ m s}^{-2}$ (2.46×10^{15}) A1
towards positive plate / upwards (and normal to plate) B1 [4]
- (ii) time = $2.4 \times 10^{-9} \text{ s}$ A1 [1]
- (d) *either* vertical displacement after acceleration for $2.4 \times 10^{-9} \text{ s}$
 $= \frac{1}{2} \times 2.46 \times 10^{15} \times (2.4 \times 10^{-9})^2$ C1
 $= 7.1 \times 10^{-3} \text{ m}$ A1
(0.71 cm < 0.75 cm and) so will pass between plates A1 [3]
i.e. valid conclusion based on a numerical value
- or $0.75 \times 10^{-2} = \frac{1}{2} \times 2.46 \times 10^{15} \times t^2$ (C1)
 t is time to travel 'half-way across' plates = $2.47 \times 10^{-9} \text{ s}$ (A1)
(2.4 ns < 2.47 ns) so will pass between plates (A1)
i.e. valid conclusion based on a numerical value
- 3 (a) mass / volume (ratio idea essential) B1 [1]
- (b) (i) mass = $Ah\rho$ B1 [1]
- (ii) pressure = force/area B1
weight (of liquid)/force (on base) = $Ah\rho g$ B1
pressure = $h\rho g$ A0 [2]
- (c) (i) ratio = 1600 or 1600:1 A1 [1]
- (ii) ratio = $\sqrt[3]{1600}$ C1
= 11.7 (allow 12) A1 [2]

Page 3	Mark Scheme	Syllabus	Paper
	GCE A/AS LEVEL – May/June 2007	9702	2

- (d) (i) density of solids and liquids are (about) equal B1 [1]
- (ii) strong forces: fixed volume B1
rigid forces: retains shape / does not flow / little deformation B1 [2]
(allow 1 mark for fixed volume, fixed shape)
- 4 (a) (i) (change in) potential energy = mgh C1
 $= 0.056 \times 9.8 \times 16$
 $= 8.78 \text{ J}$ (allow 8.8) A1 [2]
- (ii) (initial) kinetic energy = $\frac{1}{2}mv^2$ C1
 $= \frac{1}{2} \times 0.056 \times 18^2$
 $= 9.07 \text{ J}$ (allow 9.1) C1
total kinetic energy = $8.78 + 9.07 = 17.9 \text{ J}$ A1 [3]
- (b) kinetic energy = $\frac{1}{2}mv^2$
 $17.9 = \frac{1}{2} \times 0.056 \times v^2$ and $v = 25(.3) \text{ m s}^{-1}$ B1 [1]
- (c) horizontal velocity = 18 m s^{-1} B1 [1]
- (d) (i) correct shape of diagram
(two sides of right-angled triangle with correct orientation) B1
- (ii) angle = $41^\circ \rightarrow 48^\circ$ (allow trig. solution based on diagram) A2 [3]
(for angle $38^\circ \rightarrow 41^\circ$ or $48^\circ \rightarrow 51^\circ$, allow 1 mark)
- 5 (a) (i) vibrations (in plane) normal to direction of energy propagation B1 [1]
- (ii) vibrations in one direction (normal to direction of propagation) B1 [1]
- (b) (i) at (displacement) antinodes / where there are no heaps, wave has maximum amplitude (of vibration) B1
at (displacement) nodes/where there are heaps, amplitude of vibration is zero/minimum B1
dust is pushed to / settles at (displacement) nodes B1 [3]
- (ii) $2.5\lambda = 39 \text{ cm}$ C1
 $v = f\lambda$ C1
 $v = 2.14 \times 10^3 \times 15.6 \times 10^{-2}$
 $= 334 \text{ m s}^{-1}$ (allow 330, not 340) A1 [3]
- (c) Stationary wave formed by interference / superposition / overlap of B1
either wave travelling down tube and its reflection
or two waves of same (type and) frequency travelling in opposite directions B1
speed is the speed of the incident / reflected waves B1 [3]

Page 4	Mark Scheme	Syllabus	Paper
	GCE A/AS LEVEL – May/June 2007	9702	2

- 6 (a) (i) 1 total resistance = 0.16Ω A1
2 e.m.f. = *either* $(14 - E)$ *or* $(E - 14)$ A1 [2]
- (ii) *either* $14 - E = 42 \times 0.16$ *or* $(E - 14) = -42 \times 0.16$ C1
 $E = 7.3 \text{ V}$ A1 [2]
- (b) (i) charge = It C1
 $= 12.5 \times 4 \times 60 \times 60$
 $= 1.8 \times 10^5 \text{ C}$ A1 [2]
- (ii) *either* energy = EQ *or* energy = Eit C1
either energy = $14 \times 1.8 \times 10^5$ *or* energy = $14 \times 12.5 \times 4 \times 3600$
 $= 2.52 \times 10^6 \text{ J}$ A1 [2]
- (iii) energy = I^2Rt *or* Vit and $V = IR$ C1
 $= 12.5^2 \times 0.16 \times 4 \times 3600$
 $= 3.6 \times 10^5 \text{ J}$ A1 [2]
- (c) efficiency = $(2.52 \times 10^6 - 3.6 \times 10^5)/(2.52 \times 10^6)$ C1
 $= 86\%$ A1 [2]
- 7 (a) $\beta(-)$ -decay B1 [1]
- (b) $\gamma(-)$ -decay B1
either any two of Z , N and A do not change
or it is loss of energy only
or it is an electromagnetic wave B1 [2]
Allow ' $\alpha(-)$ -decay' as change of 4 in the nucleon number cannot be shown on the diagram' (B2)
Do not give credit for a 'bald' $\alpha(-)$ -decay)